Nicholas Kadunce

OPERATIONS LEADER · AI & DIGITAL TRANSFORMATION · P&L EXECUTIVE

MANUFACTURING EXCELLENCE | AI-POWERED SYSTEMS | FULL-STACK DEVELOPMENT | HIGH-PERFORMANCE TEAMS

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Proven operations leader who builds high-performing teams, architects AI systems that multiply operational capacity, and delivers measurable P&L results. Designed and deployed enterprise AI solutions—from real-time production intelligence (JMOS: 16 machines, 2.8M+ sensor records) to automated reporting, intelligent process documentation, and workflow automation that eliminates manual effort across entire organizations. Skilled at translating complex operational challenges into scalable digital solutions, shaping strategy, and aligning execution with long-term growth.

CORE COMPETENCIES

Leadership & Strategy
High-Performance Team Building · Change Leadership · P&L Accountability · Talent Development · Cross-Functional Leadership · Organizational Design

AI & Digital Transformation
AI-Powered Operations Intelligence · Enterprise Workflow Automation · Automated OEE & SPC Analytics · AI Agent Development · Intelligent Process Documentation · Full-Stack Dev (Node.js, PostgreSQL, MySQL)

Operational Excellence
Lean & Six Sigma · Manufacturing Optimization · Risk Management · Quality Management Systems (ISO 9001) · CMMS Implementation · Supplier & Inventory Management

Strategic Planning & Metrics
KPI Development & Dashboarding · Production Scalability · Financial Forecasting & Modeling · Data-Driven Decision Making · Project & Program Management

PROFESSIONAL EXPERIENCE

JENNMAR
2025 – Present

Plant Manager — Underground Mining & Tunneling Infrastructure Solutions
Reedsville, WV

Lead end-to-end plant operations accountable for safety, quality, production, and financial performance. Architect AI-powered digital systems that automate data collection, surface real-time insights, and replace manual processes with intelligent workflows.

AI & DIGITAL TRANSFORMATION

- Designed and deployed **JMOS**, a full-stack AI production monitoring platform (Node.js, Express, PostgreSQL, MySQL) integrating real-time IoT sensor feeds from 16 machines, automated OEE computation, intelligent downtime detection, and operator-facing classification workflows — eliminating manual data entry plant-wide.
- Engineered an automated downtime intelligence engine ingesting 2.8M+ sensor records, applying statistical cycle-time analysis to distinguish true downtime, and surfacing Pareto-ranked root causes for targeted improvement.
- Built a real-time caching and data pipeline layer with 60-second refresh cycles, enabling instant dashboard response across all clients while processing ~15K daily sensor readings.
- Developed an AI-assisted Best Demonstrated Rate (BDR) system that statistically detects operator performance gains, auto-generates rate-change proposals with full OEE context, and routes approvals to management.
- Implemented automated SPC and defect analytics with real-time control charting, Pareto analysis, and trend detection for proactive quality interventions.
- Created AI-automated daily Operations Performance Reports rendered as SVG visualizations with OEE breakdowns and equipment-level KPIs — auto-distributed to leadership each morning.
- Architecting enterprise-wide AI deployment roadmap — automated SOP standardization across facilities, AI-driven DMAIC automation, cross-training intelligence, and voice-based operational logging to eliminate indirect labor and human error.

OPERATIONAL LEADERSHIP

- Own full P&L for a high-volume manufacturing site, driving **36.7% output increase** within first year through operational cadence redesign and data-driven target setting.
- Built a financial forecasting model that improved operating net income visibility and reduced month-end variance, enabling faster executive decision-making.
- Designed and installed a cross-functional daily management system aligning production, maintenance, quality, and safety — eliminating siloed reporting and accelerating issue resolution.
- Led strategic shift from reactive to planned maintenance through CMMS implementation, reducing unplanned downtime and extending asset life across all critical equipment.
- Developed and coached front-line leadership through structured Supervisor Standard Work, raising accountability, reducing turnover, and building a pipeline for internal promotion.

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PROFESSIONAL EXPERIENCE (CONTINUED)

EOS ENERGY ENTERPRISES, INC.

2023 – 2025

Operations Manager — Battery Energy Storage Systems (BESS) & Module Operations

Turtle Creek, PA

Transformed manufacturing operations through data-driven systems, high-performing teams, and continuous improvement to achieve measurable results in efficiency, quality, and production scalability.

MODULE OPERATIONS (700 BUILDING)

- Built the company's first production dashboards with OEE metrics, automated gap analysis, and KPI control charts enabling real-time visibility and strategic decision-making.
- Transitioned subassembly from batch to one-piece flow, **increasing production 41%** and reducing defects by 22%.
- Reduced equipment downtime by 36% within 2 months through data-driven Pareto analysis and mitigation plans.
- Led and trained a 40-person third-shift team from scratch — became **the highest-performing shift within one month**.

BESS OPERATIONS (200 BUILDING)

- Increased daily BESS production from 0.5 to 2.2 units/day via Yamazumi workload balancing, Takt time alignment, and team-based workflows — a **340% increase**.
- Reduced rework by 66% through data collection, Pareto analysis, and targeted process improvements.
- Integrated an Andon system with automated data collection and control charts for real-time monitoring of quality, downtime, and production metrics.

Quality Engineer — Battery Energy Storage Systems & Module Operations

- Engineered a high-efficiency battery testing routine, **reducing cycle times by 60%** and improving overall production throughput.
- Led defect reduction initiatives achieving a 30% decrease in scrap defects and reducing downstream quality excursions.
- Obtained ISO 9001 Lead Auditor licensure and spearheaded internal audits aligning operations with international standards.

MESSER AMERICAS

2022 – 2023

Regional Production Engineer — ASU Bulk, Tonnage, & Specialty Industrial Gases

Northeast Region, USA

- Led Six Sigma initiatives boosting Krypton/Xenon production by 15%, **increasing annual revenue by over $1 million**.
- Drove 15+ system enhancement projects, cutting annual downtime by 25 days and improving daily output consistency by 16%.
- Streamlined production and delivery processes through cross-functional collaboration, reducing average lead times by 30 days.

JOHNSON MATTHEY

2020 – 2022

Production Team Shift Leader II / Process Control Engineer — Clean Air Division

Smithfield, PA

- Increased OEE by over 5% in 2022, achieving **zero lost-time and recordable injuries** across all shifts.
- Led process optimization achieving an 8% cycle time reduction and a 4% decrease in defects.
- Designed a risk management tool, reducing LTY by 25% and downtime by 8 minutes per Work Order.

CERTIFICATIONS & EDUCATION

- **Certified Lean Six Sigma Black Belt (CSSBB)**
  Six Sigma Global Institute (SSGI)
- **Certified Lean Six Sigma Green Belt (CSSGB)**
  Messer Americas, EMBRACE Training Academy
- **ISO 9001 Lead Auditor**
  Exemplar Global
- **Certified ScrumMaster® (CSM)**
  Scrum Alliance

B.S., Chemical Engineering

West Virginia University — Morgantown, WV

Class of 2020